

Gan 2026 Full Research Retrospective

A print-ready synthesis of six architecture families, what each one taught us, and how close the program is to its core research thesis.

BEST TRANSPARENT COMPARATOR Rules-only V1 reaches 0.9293 Purist on validation, but falls to about 0.7600 on locked test.
CORE ATTRIBUTION LESSON The 0.9000 structured-events validation result is repair-heavy hybrid behavior, not clean LLM-first success.
CURRENT RESEARCH PATH The state graph now separates coverage, projection, boundary nodes, invariance, and arbitration.

Scope: current Gan 2026 seizure-frequency research state across `rules_only_v1`, `llm_only_structured_events`, `llm_only_claim_table_selector`, `hybrid_rules_candidates_llm_adjudicator`, `hybrid_clinical_frequency_state_graph`, and `llm_heavy_clinical_frequency_reasoner`.

This is a research retrospective, not a benchmark-comparison claim. Validation results are development evidence under `gan2026_split_v1`; locked-test results are aggregate/frozen-audit context only and must not be used for row-level tuning.

Executive Synthesis

The project is materially closer to its core research goals, but not because one architecture has simply "solved" Gan 2026. The clearest finding is that seizure frequency extraction is a multi-stage clinical state problem: source evidence, temporality, assertion, seizure-free intervals, clusters, diary windows, competing semiologies, underdetermined frequency, and benchmark-facing label grammar all have to be represented separately before the final Gan label is safe.

The work has produced three kinds of progress:

1. A strong transparent comparator: `rules_only_v1` reaches 697/750 = 0.9293 Purist and 704/750 = 0.9387 Pragmatic on validation, with exact selected evidence on every row. It also drops to about 0.7600 Purist and 0.7867 Pragmatic on locked test, showing that validation strength alone is not a generalisation claim.
2. A disciplined attribution story: structured LLM and claim-table experiments showed useful source-near extraction, but also exposed attribution drift. The threshold-passing structured-events result, 675/750 = 0.9000 Purist, is repair-heavy hybrid behavior rather than clean LLM-first behavior.
3. A better research substrate: the state-graph cycle separates coverage, projection, boundary-state node construction, invariance, and arbitration. Validation50 oracle coverage is 47/50 and projection Purist/Pragmatic F1 is 0.9600; validation hard-slice oracle coverage is 219/250 with projection Purist F1 0.9160. The system is now set up to ask sharper component questions instead of chasing another broad aggregate validation score.

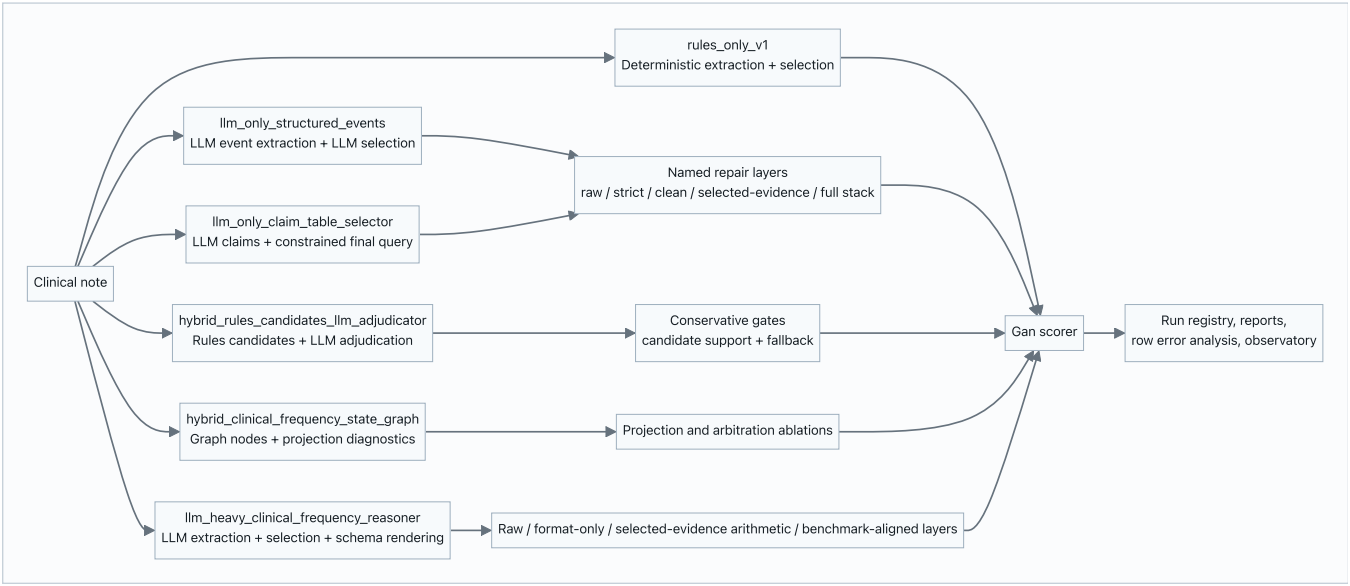
The main negative result is equally important: changing the surface architecture does not by itself remove the generalization gap. Rules-only, claim-table, structured-events, and hybrid adjudicator variants all encounter hidden subfamilies in template, temporal anchoring, cluster notation, seizure-free boundary language, and distributed counts. The next research leap is not another prompt patch; it is proving which component owns each clinical decision and whether that ownership transfers beyond the validation surface.

Core Research Goals Revisited

The contribution thesis asks for a modular, auditable clinical extraction system where deterministic rules and LLM reasoning are explicit, testable components. Against that thesis, we are in a stronger position than the raw metrics alone suggest.

GOAL	CURRENT STATE	PROGRESS
Modular breadth and depth	The Gan 2026 package is organized into <code>contract</code> , <code>deterministic</code> , <code>selected_evidence</code> , <code>llm</code> , <code>hybrid</code> , <code>state_graph</code> , <code>reports</code> , <code>experiments</code> , <code>artifact_analysis</code> , <code>observatory</code> , and <code>cli</code> ownership areas.	Strong. The seizure-frequency task is no longer a pile of one-off scripts.
Generalisation by design	Split protocol is locked; validation/test distinction is explicit; saturated validation has pushed work toward hard slices, synthetic panels, and frozen audits.	Medium-strong. We have evidence of the gap and better tools to study it, but not yet a generalizing final model.
Transparency through evidence and error analysis	Every major family now emits score layers, evidence validity summaries, repair metadata, and row/error artifacts. State graph exposes representability and projection separately.	Strong. This may be the strongest achieved contribution so far.
Deterministic rules as controlled variables	Deterministic V1 is frozen; rule families are catalogued and ablated; post-LLM repair is now named rather than hidden.	Strong, with a caveat: some early LLM repair work became too semantic before the audit forced a cleaner taxonomy.
LLM reasoning as an explicit component	Claim-table, structured-events, adjudicator, boundary-node builder, and LLM-heavy reasoner all test different LLM roles.	Medium. LLMs are useful for evidence/claim generation, but clean LLM-owned final-label performance remains below the target.

Architecture Map



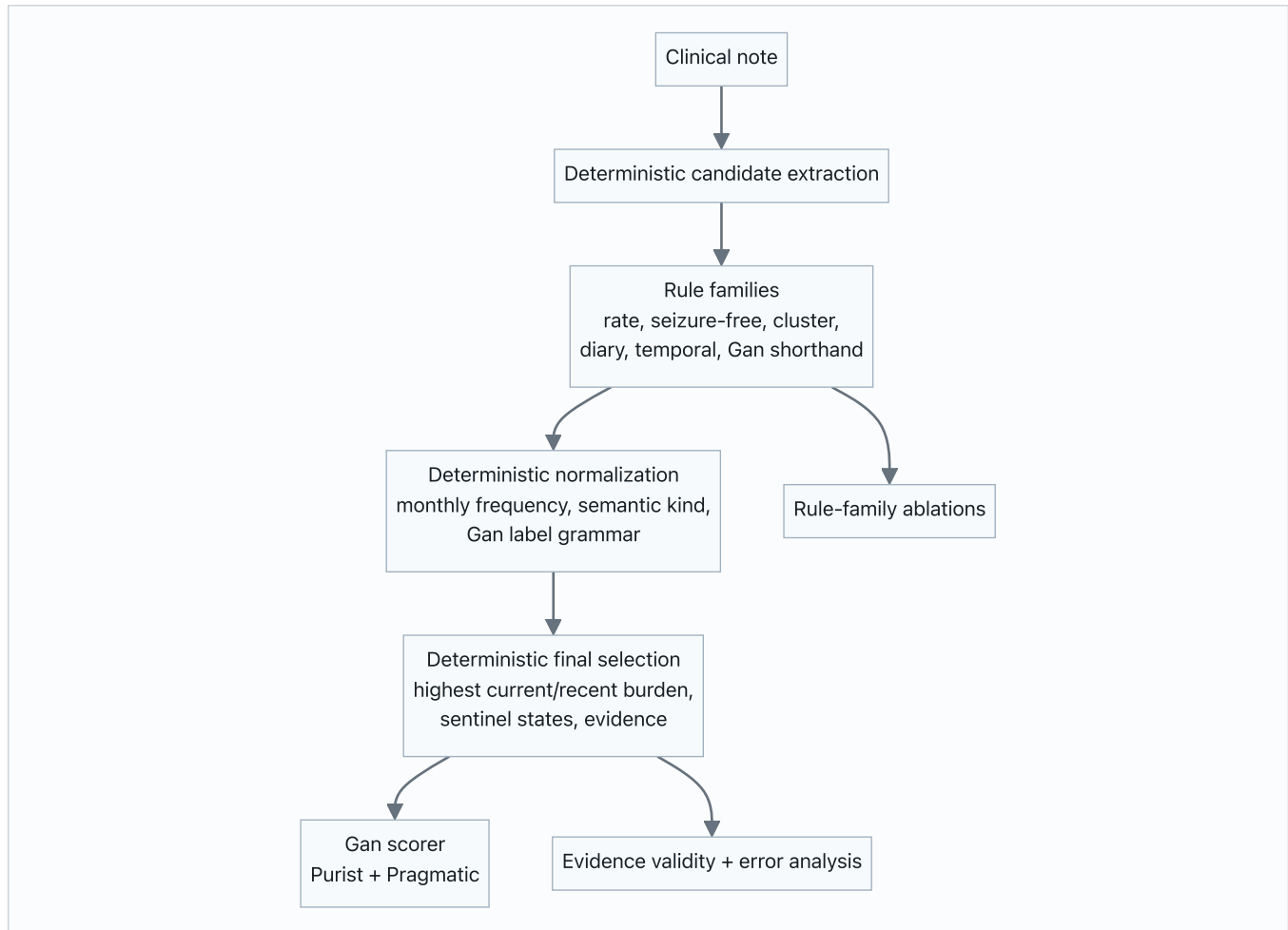
The most important architectural lesson is that a final Gan label is too lossy to be the only representation. Every successful or instructive branch has moved toward richer intermediate objects: candidate events, claim tables, adjudication records, graph nodes, selected evidence, and repair-mode metadata.

Performance Summary

FAMILY	BEST/CURRENT SURFACE	RESULT	INTERPRETATION
rules_only_v1	validation750	697/750 Purist = 0.9293; 704/750 Pragmatic = 0.9387	Strongest transparent validation comparator.
rules_only_v1	locked test450	about 343/450 Purist = 0.7622 by rerun; original report 0.7600; 354/450 Pragmatic = 0.7867	Generalization gap is decisive.
llm_only_structured_events v0.5 full stack	validation750	675/750 Purist = 0.9000; 690/750 Pragmatic = 0.9200	Hits threshold only as repair-heavy hybrid behavior.
Structured-events clean attribution	650-row saved-output replay	raw 394/650 = 0.6062; clean 438/650 = 0.6738	Clean LLM-first endpoint remains well below target.
llm_only_claim_table_selector v4	validation250 then validation750	231/250 clean Purist, then 528/750 clean Purist	Prefix optimism; full-validation collapse.
llm_only_claim_table_selector v5	validation250 then test450	227/250 clean Purist; 301/450 clean Purist	Useful complementarity, not reliable replacement.
hybrid_rules_candidates_llm_adjudicator v0.1	validation750	680/750 Purist, 689/750 Pragmatic	Good decomposition; LLM regressed more deterministic-correct rows than it fixed.
Hybrid v0.2 cluster_diary_candidate_recall	validation750	677/750 Purist; 686/750 Pragmatic	Underperformed deterministic top despite stronger hard-case recall.
Hybrid v0.2 cluster_diary_candidate_recall	locked test450 frozen audit	343/450 Purist; 353/450 Pragmatic	Tied deterministic Purist, slightly worse Pragmatic; no promotion.
hybrid_clinical_frequency_state_graph diagnostics	validation50	47/50 oracle coverage; 0.9600 projection Purist/Pragmatic F1	Promising diagnostic substrate, not a final benchmark result.
State graph hard-slice union	validation hard-slice 250	219/250 oracle coverage; 0.9160 projection Purist F1	Coverage and projection are now measurable separately.
llm_heavy_clinical_frequency_reasoner v0	validation25 smoke	24/25 schema-valid; 18/25 selected evidence exact; raw scorable 0/25; selected-evidence arithmetic 23/25	Useful evidence selection, but fails LLM-heavy Stage A.
llm_heavy_clinical_frequency_reasoner v1	validation50 then validation250	validation50: 50/50 structured, 48/50 selected evidence exact, raw Purist 41/50; validation250: 237/250 structured, raw Purist 188/250, selected-evidence arithmetic 219/250	Output contract improved; validation250 rejects promotion because the best layer remains attribution-diagnostic.

1. rules_only_v1

Architecture



Why This Architecture

The initial rules-only system was the fastest way to make the task concrete. It forced the repo to define label parsing, source evidence, row quality, normalization, final selection, and scoring contracts. It also made deterministic logic scientifically inspectable rather than incidental.

The rule families now have research meaning:

- portable rate expressions;
- seizure-free/no-event assertions;
- cluster arithmetic;
- diary/log aggregation;
- temporal selection;
- Gan shorthand;
- benchmark-facing format repair.

What We Did

We built deterministic extraction and final selection, then froze it as `rules_only_v1`. We ran validation, locked holdout, error analysis, and rule-family ablations. The rule catalogue and architecture docs now treat rules as categorized, testable components.

Key validation ablation findings:

DISABLED GROUP	PURIST	DELTA
none	0.9293	baseline
portable rate expressions	0.7627	-0.1666
temporal selection	0.7787	-0.1506
seizure-free/no-event assertions	0.8107	-0.1186
diary log aggregation	0.8507	-0.0786
cluster arithmetic	0.8600	-0.0693
Gan shorthand	0.9027	-0.0266
benchmark repair	0.9293	0.0000

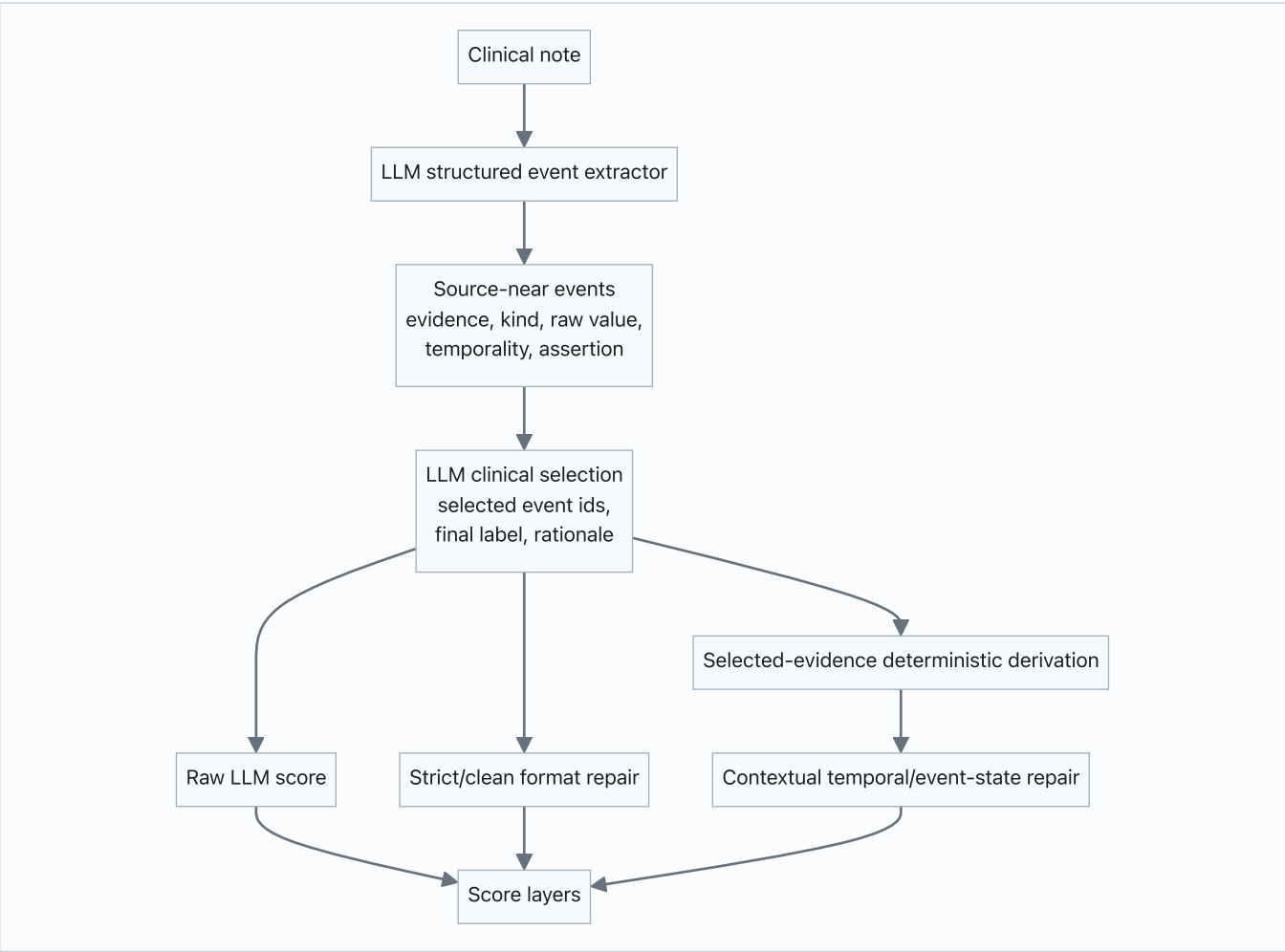
Performance And Lessons

The system is excellent on validation and perfectly evidence-exact on both validation and test. But the locked-test drop is the defining warning. Evidence exactness is necessary but not sufficient: the hard operation is converting a span into a current clinical state under multiple windows, semiologies, clusters, and boundary states.

Rules-only taught us that high validation F1 can be achieved transparently, but it also taught us that a local rule stack can become brittle. The right role for `rules_only_v1` is frozen comparator, diagnostic substrate, and ablation control, not endlessly tuned production policy.

2. llm_only_structured_events

Architecture



Why This Architecture

The direct note-to-label approach asked too much of one model pass. Structured events were introduced to preserve source-near clinical facts before final selection. This matched the thesis: store intermediate events, not just final predictions.

What We Did

We moved from direct label prompting to a slim event schema, added a general LLM experiment CLI, used cache-first runs and saved-output replay, separated schema-repair from normalization, and eventually built attribution ladders for raw, strict, clean, selected-evidence-derived, and full-stack scoring.

The turning point was the repair audit. The 750-row v0.5 completion reached the 0.9000 validation target, but it carried 481 deterministic repair notes and 714/750 exact selected-evidence substrings. The grouped 650-row ladder clarified where performance came from:

LAYER GROUP	PURIST	INTERPRETATION
Raw structured LLM selection	394/650 = 0.6062	Model's own final label.
Clean scorer-facing normalization	438/650 = 0.6738	Clean LLM-first endpoint.
Broad basic label repair bridge	461/650 = 0.7092	Crosses into semantic repair.
Selected-evidence deterministic derivation	546/650 = 0.8400	Main metric jump.
Contextual temporal/event-state modules	588/650 = 0.9046	Full hybrid stack.

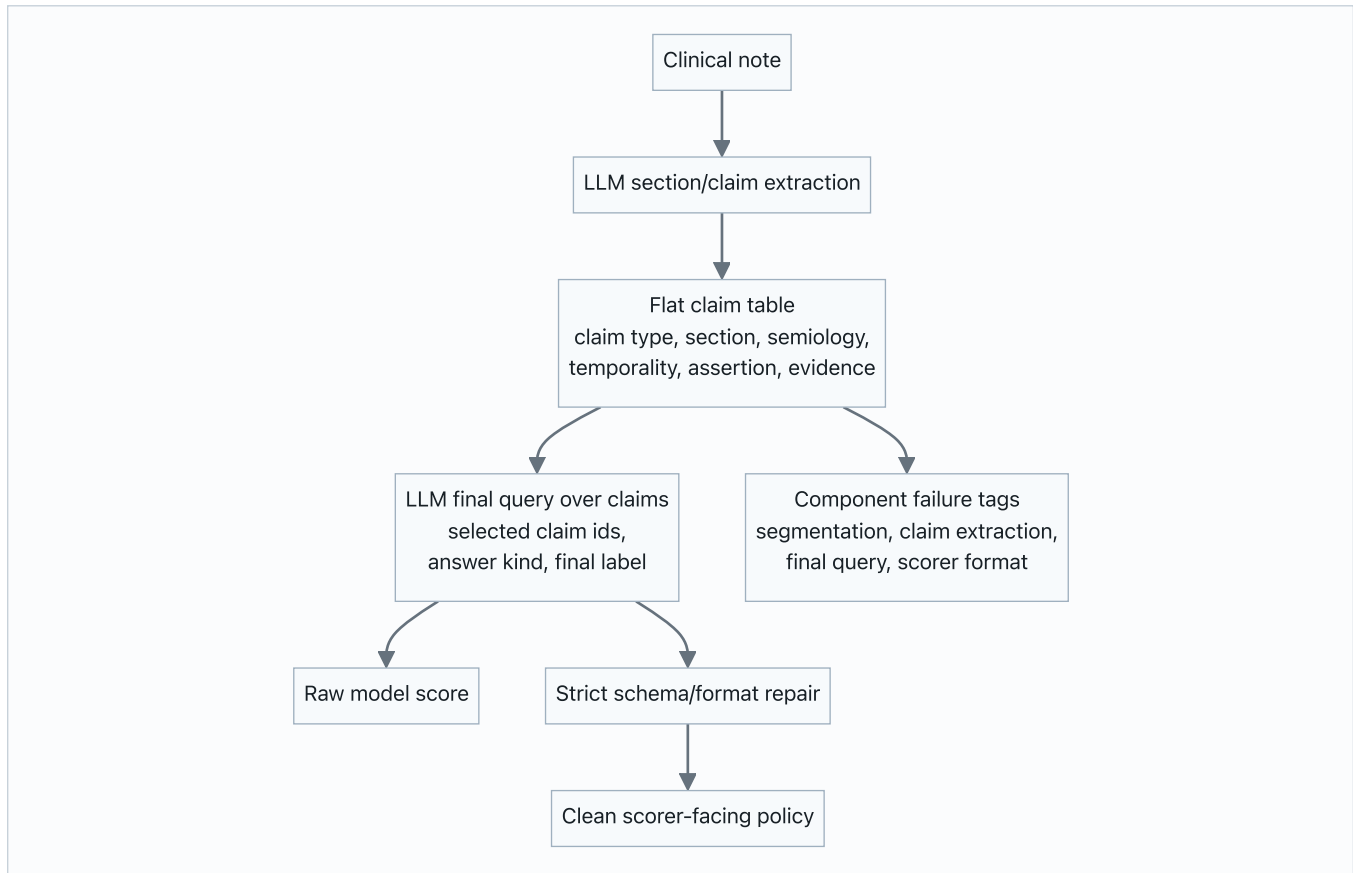
Performance And Lessons

Structured-events is valuable because the LLM often finds useful evidence and intermediate events. It is not yet strong as a clean LLM-first final-label system. The largest performance movement came after deterministic derivation over model-selected evidence, which means the high score is a hybrid result.

The central lesson is attribution discipline: selected evidence is not a magic safe boundary. If deterministic code derives a better label from the evidence than the model emitted, that deterministic derivation is prediction-bearing and must be named, ablated, and claimed honestly.

3. `llm_only_claim_table_selector`

Architecture



Why This Architecture

The claim-table branch tested a more LLM-native representation than event normalization. Instead of asking the model to directly solve arithmetic and selection inside one final answer, it first produced a source-near table of claims, then answered from that table. This made segmentation, claim extraction, temporality, final query, and scorer formatting separately inspectable.

What We Did

We developed v3/v4 claim tables, then v5 with more explicit constrained selector state, cluster-axis fields, and boundary-state fields. We added component ablation and component-failure reports. The architecture initially looked strong on validation prefixes:

- v4 clean validation250: 231/250 = 0.9240 Purist.
- v5 clean validation250: 227/250 = 0.9080 Purist.

But broad surfaces exposed the instability:

- v4 clean validation750: 528/750 = 0.7040 Purist and 577/750 = 0.7693 Pragmatic.
- v5 clean test450: 301/450 = 0.6689 Purist, with 445/450 structured records.

Performance And Lessons

Claim tables are transparent and complementary, but not yet reliable. On test, v5 correctly classified 61 rows that deterministic V1 missed, while deterministic V1 correctly classified 103 rows that v5 missed. That is a strong signal for ensemble or hybrid arbitration, but a weak signal for replacement.

The failure modes were meaningful:

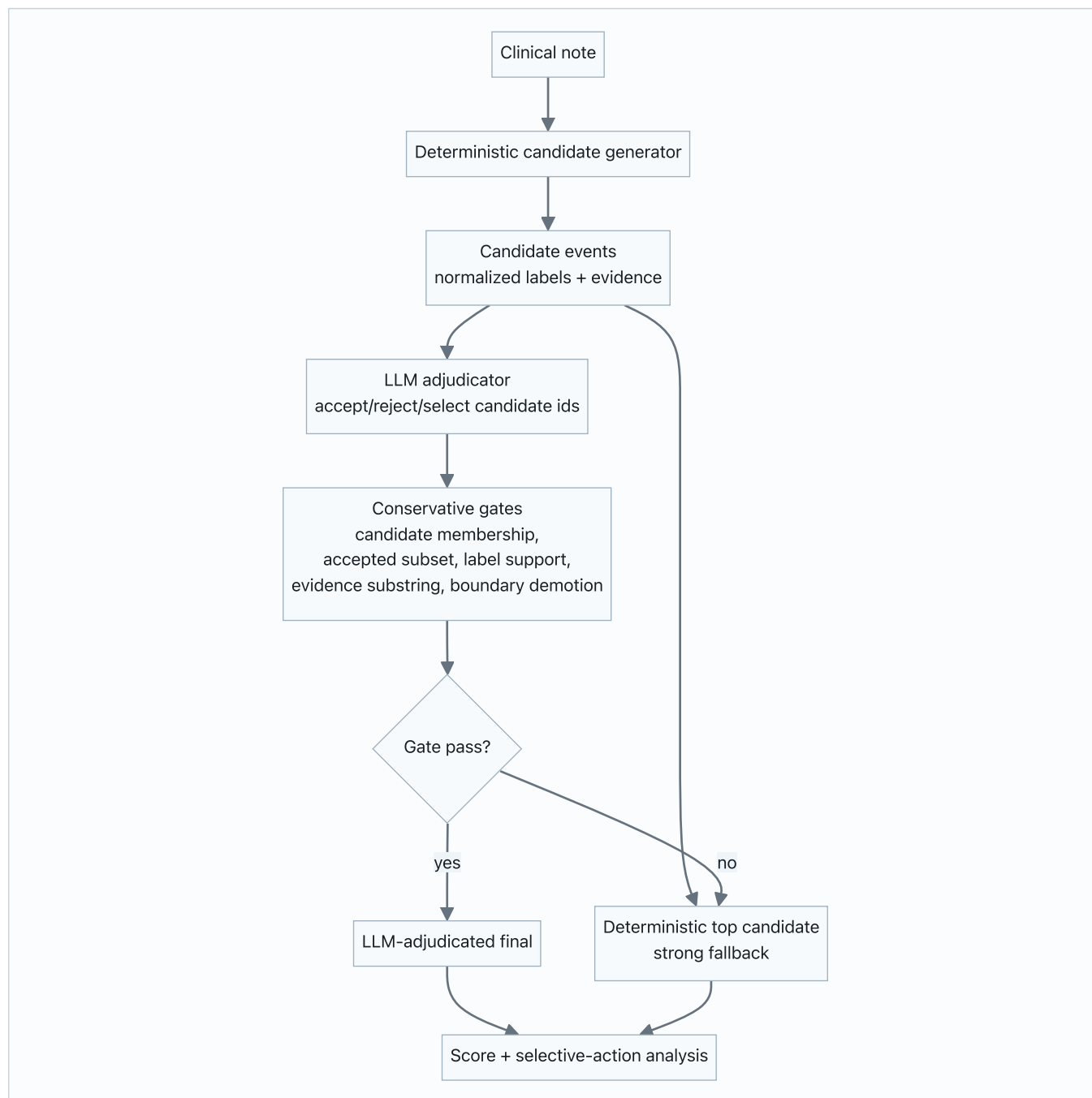
- claim extraction failures rose from 9/250 on validation250 to 61/450 on test;

- final-query failures rose from 4/250 to 32/450;
- scorer-format failures rose from 8/250 to 39/450;
- selected final evidence exactness dropped from 246/250 to 418/450.

The lesson is that source-near claim tables help transparency, but the final Gan query still collapses too much clinical state unless cluster axes, boundary-state decisions, and denominator policy are handled explicitly.

4. hybrid_rules_candidates_llm_adjudicator

Architecture



Why This Architecture

This family tested a pragmatic hybrid thesis: deterministic rules are good at high-recall candidate retrieval, while an LLM may be better at semantic adjudication among candidates. The v0.2 design made deterministic top the fallback and allowed LLM changes only through named overreach gates.

What We Did

We first ran v0.1 over deterministic candidates. It cleared broad validation numerically after schema replay, but underperformed deterministic top:

- validation750 LLM adjudicator: 680/750 = 0.9067 Purist;
- deterministic top on same rows: 697/750 = 0.9293;
- LLM corrected 7 deterministic misses but regressed 24 deterministic-correct rows.

We then built v0.2 conservative gates and saturated-surface analysis. The v0.2 validation250 result was not a promotion:

- deterministic top: 246/250 Purist;
- gated adjudicator: 244/250 Purist;
- deterministic-wrong to gated-correct: 0;
- deterministic-correct to gated-wrong: 2.

Synthetic hard-case stress then identified a candidate-recall problem. The named `cluster_diary_candidate_recall` revision improved the 56-row synthetic panel:

- baseline synthetic gated Purist: 42/56;
- cluster/diary revision gated Purist: 50/56;
- raw wrong-to-correct: 13;
- raw correct-to-wrong: 0.

We froze that revision for a comparator-only generalization audit.

Performance And Lessons

The broad audit showed the revision did not generalize:

SURFACE	CONDITION	PURIST	PRAGMATIC	CHANGED LABELS	WRONG TO CORRECT	CORRECT TO WRONG
validation750	deterministic top	697/750	704/750	0	0	0
validation750	gated final	677/750	686/750	45	5	25
test450	deterministic top	343/450	354/450	0	0	0
test450	gated final	343/450	353/450	29	9	9

Candidate recall became the hard ceiling:

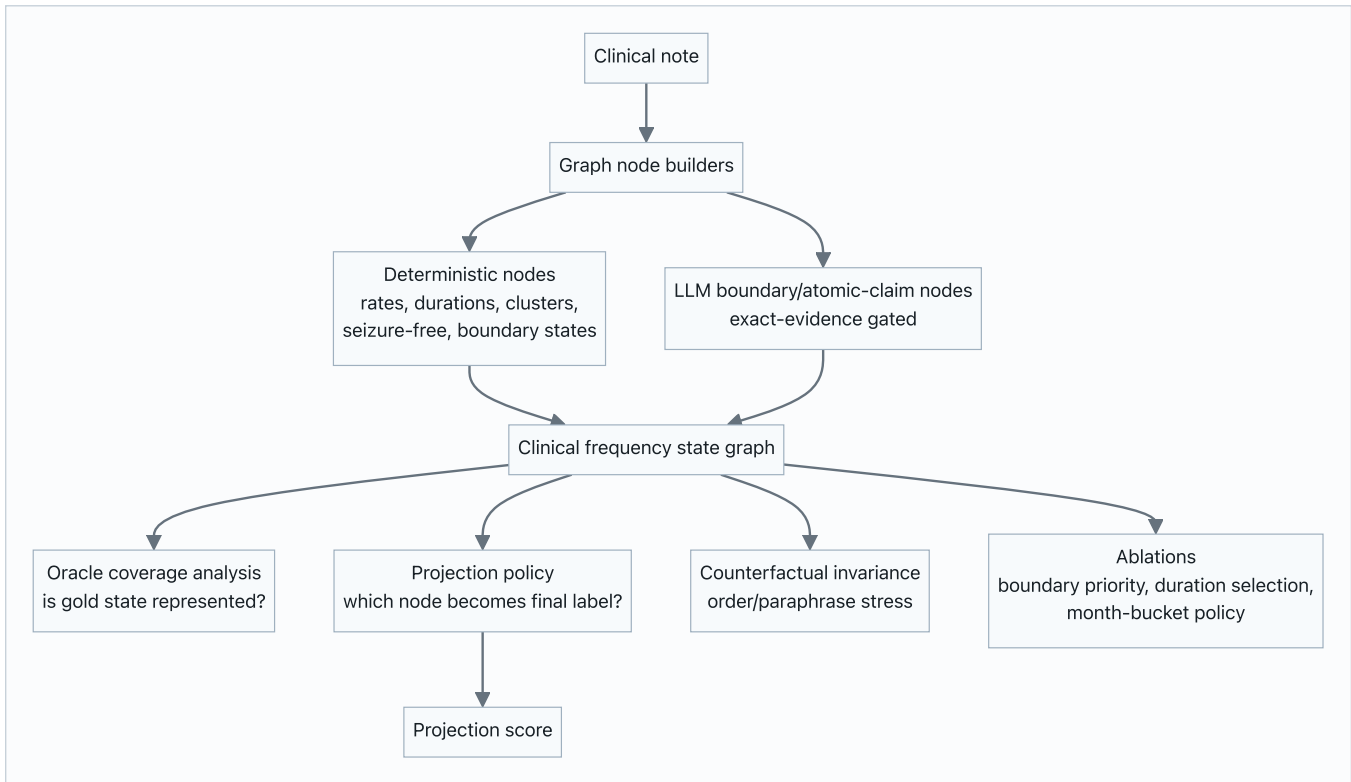
SPLIT	CANDIDATE PURIST RECALL	NO-RECALL ROWS	NO-RECALL ROWS FIXED BY GATED FINAL
validation750	707/750 = 0.9427	43	0
test450	359/450 = 0.7978	91	0

The lesson is sharp: an adjudicator constrained to a candidate set cannot recover rows where the correct category is absent. The LLM did contain signal, especially raw test changes before gates, but current gates are blunt and candidate recall is too low on shifted surfaces.

This result moved the project away from final-label prompt/gate tuning and toward semantic-state graph diagnostics.

5. hybrid_clinical_frequency_state_graph

Architecture



Why This Architecture

The state graph is the most important architectural reframing so far. It stops treating "did we emit the right final label?" as the only question. Instead, it asks:

- Did the system represent the correct clinical state somewhere?
- Is the selected evidence exact?
- Did projection select the right represented state?
- Which boundary states are missing?
- Is the graph invariant to harmless note-order or paraphrase perturbations?

That separation directly supports the paper thesis. It gives us a substrate for coverage, projection, and arbitration ablations, and it prevents node construction improvements from being confused with scorer-policy changes.

What We Did

We scaffolded deterministic graph diagnostics, reused saved LLM atomic-claim outputs where appropriate, ran validation and synthetic diagnostics, reviewed row families, built a hosted boundary-state graph-builder for unknown and unresolved-multiple nodes, replayed accepted nodes, and ran projection arbitration plus seizure-free duration ablations.

Key results:

- validation50 oracle coverage: 47/50;
- validation50 projection Purist/Pragmatic F1: 0.9600;
- validation hard-slice union oracle coverage: 219/250 = 0.876;
- validation hard-slice projection Purist F1: 0.9160;
- counterfactual order invariance: 1.0;
- counterfactual paraphrase invariance: 0.98;
- LLM atomic-claim conversion: 79/80 exact-evidence-certain nodes, with one non-exact claim downgraded.

The boundary-state builder then recovered useful missing coverage:

- validation31 produced 10/31 representability-gain candidates;
- synthetic unknown8 v1 produced 8/8 schema-valid, exact-evidence-valid unknown nodes;
- accepted-node replay recovered representability on all 10 validation gain rows, but unchanged projection exactly matched only 6/10 labels.

Seizure-free duration analysis separated node construction from projection:

- duration-node replay emitted 21/21 exact-evidence-valid nodes over 18 target rows;
- month-scale representability recovered on all 18 target rows;
- unchanged projection still recovered 0/18 exact duration labels.

Projection then became the bottleneck. The enriched projection replay found `month_bucket_duration_selection` could recover 18/18 exact duration labels on the target surface. A broader v0 ablation fixed those 18 target rows but caused 27 already-correct regressions. Gated v1 preserved the 18/18 target corrections and reduced regression-panel changes to 4/232, with 0 already-correct and 0 frequency-with-seizure-free regressions. It remains revise-only pending a broader enriched-node regression panel.

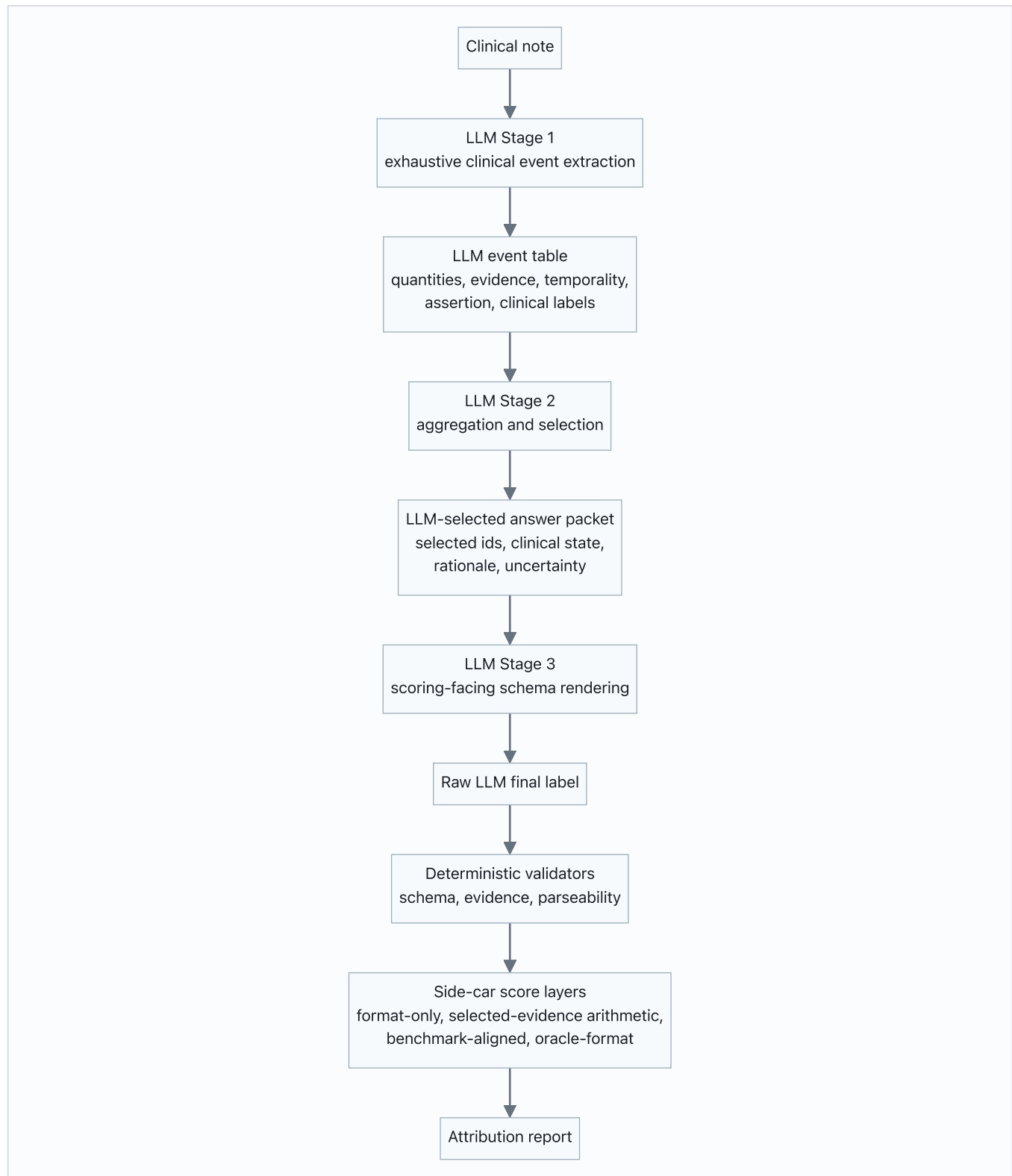
Performance And Lessons

The state graph has not yet produced a promoted final production policy. That is the right decision. Its contribution is stronger: it has localized the problem. Coverage is promising; boundary nodes can be added with exact evidence; duration-node construction can be fixed; projection and arbitration remain the active bottleneck.

This family moves us closest to the core research goals because it keeps clinical meaning rich until the final projection step. It also gives paper-grade tables: oracle coverage, projection-only F1, invariance, boundary-node recall, duration-node ablations, and regression panels.

6. `llm_heavy_clinical_frequency_reasoner`

Architecture



Why This Architecture

After the state-graph cycle became mostly deterministic, we needed a strong LLM-driven alternative. The LLM-heavy protocol deliberately flips ownership: the model owns extraction, normalization proposal, aggregation/selection, and final schema rendering. Deterministic code validates, scores, and optionally applies named benchmark-alignment adapters. If deterministic code changes the selected event or clinical state, the result

is no longer LLM-heavy.

What We Did

We wrote the protocol, implemented and registered `llm_heavy_clinical_frequency_reasoner_v0`, ran validation25 with GPT-4.1 mini, then performed saved-output schema replay and row-level error analysis. After that error analysis, we implemented v1 with a stricter output contract:

- `final_answer.selected_evidence` must equal one selected event evidence string rather than a concatenated rationale;
- prose clinical interpretation moves into `raw_clinical_summary`, `combined_rationale`, or `final_rationale`;
- `raw_llm_final_label` must be parser-ready, with explicit examples for inequalities, upper bounds, cluster modifiers, and seizure-free distractors;
- shape-only parser repair covers singleton enum lists and non-semantic aliases such as `vague_count=many`, `vague_count=several`, `vague_count=most days`, unsupported quantity units such as `hour`, and `raw_llm_final_kind` aliases such as `cluster_frequency -> frequency`.

Aggregate validation25 results:

LAYER	SCORABLE	PURIST CORRECT	PRAGMATIC CORRECT
raw LLM	0/25	0/25	0/25
format-only	11/25	10/25	10/25
selected-evidence arithmetic	24/25	23/25	23/25
benchmark-aligned	24/25	13/25	13/25
oracle-format upper bound	11/25	10/25	10/25

Additional contract metrics:

- structured records after non-semantic schema replay: 24/25;
- event evidence exact: 42/47;
- selected evidence exact: 18/25;
- selected-event trace mismatches: 0/25;
- frozen deterministic V1 same-row Purist correct: 25/25.

The v1 validation ladder then produced:

SURFACE	STRUCTURED	PARSE/SCHEMA FAILURES	SELECTED EVIDENCE EXACT	RAW/FORMAT-ONLY PURIST	SELECTED-EVIDENCE ARITHMETIC PURIST	BENCHMARK-ALIGNED PURIST
validation50	50/50	0	48/50	41/50	48/50	45/50
validation250	237/250	13	230/250	188/250	219/250	204/250

Validation50 passed the output-contract gate and justified a diagnostic validation250 scale-up. Validation250 rejected promotion: the raw prediction-bearing LLM label remained far below the target, while the stronger selected-evidence arithmetic score still depended on deterministic derivation over model-selected evidence.

Performance And Lessons

The v0 reasoner fails the Stage A smoke gate. The model often identifies the right evidence and selected event trace, but its raw final labels are not parser-ready and final selected evidence is often a concatenation or paraphrase rather than one exact source substring. V1 fixes much of this contract problem: the 50-row gate was schema-clean and raw labels were usually scorable.

The high selected-evidence arithmetic score is promising but attribution-invalid for LLM-heavy promotion: deterministic arithmetic over selected evidence is doing the scoring work. The benchmark-alignment adapter also regressed 10 rows relative to selected-evidence arithmetic, showing that broad Gan repair over raw prose is unsafe for this architecture.

The validation250 result narrows the next revision target. It is no longer enough to ask for parser-ready labels and exact selected evidence: v1 must be audited for parse/schema tail failures, selected-event trace mismatches, and raw-label mistakes relative to selected-evidence arithmetic. The evidence selection signal remains useful, but the architecture has not yet shown that the model can own final Gan label rendering cleanly at scale.

Cross-Architecture Findings

1. Evidence Exactness Is Necessary But Not Sufficient

`rules_only_v1` has exact evidence on every validation and test row. Claim-table and structured-events variants often have high evidence exactness too. Yet all families show either validation-to-test gaps or clean-attribution weakness. The hard part is not just finding text; it is choosing the correct clinical state and benchmark-facing abstraction.

2. Candidate Recall Defines Hybrid Headroom

The hybrid adjudicator cannot fix rows where the correct candidate is absent. On test, candidate Purist recall fell to $359/450 = 0.7978$; every no-recall row remained wrong after gated final. This makes candidate generation a scientific object, not mere preprocessing.

3. Repair Layers Can Hide Prediction-Bearing Logic

Structured-events reached 0.9000 validation only after deterministic repair families that changed semantic answers. The project learned to report raw, strict, clean, selected-evidence-derived, benchmark-aligned, and full-stack layers separately. This is now one of the strongest methodological safeguards in the repo.

4. Validation Prefixes Are Often Too Friendly

Claim-table v4 went from 231/250 clean Purist to 528/750 clean Purist. Structured-events v0.5 dropped from a very high validation250 replay to exactly 675/750 on full validation. Hybrid v0.2 looked strong on validation250 but did not beat deterministic top on validation750 or test450. Prefixes are useful development gates, not promotion proof.

5. Hidden Subfamilies Matter More Than Top-Level Label Mix

Validation and test label distributions are not wildly different at the top level. The shift is inside categories: note template, temporal anchor, cluster phrasing, diary arithmetic, seizure-free boundary language, vague frequency claims, and competing semiologies. Future splits and hard panels should stratify by these subfamilies, not only by gold kind.

6. LLMs Are Useful, But Their Best Current Role Is Not Settled

The evidence supports several possible LLM roles:

- source-near event extractor;
- claim-table builder;
- candidate adjudicator;
- boundary-state graph-node builder;
- LLM-heavy clinical reasoner.

So far, the strongest clean LLM-owned final-label result is not competitive with the deterministic comparator. The most reliable LLM signal appears in source-near extraction, selected evidence, claim decomposition, and targeted boundary-node construction. The LLM-heavy track remains necessary because it tests whether stronger prompting/schema discipline can make the model own more of the clinical reasoning without hidden repair.

How Much Closer Are We?

We are substantially closer to the research paper, but only partially closer to a final generalizing system.

Closer To The Paper

Very close. We now have:

- a frozen rules-only comparator with ablations;
- multiple LLM/hybrid architectures with named score layers;
- a documented attribution failure and repair taxonomy;
- a generalization-gap report across validation/test;

- state-graph diagnostics that separate coverage from projection;
- evidence-validity and schema-validity metrics;
- a run registry and observatory direction;
- conservative claim language around test and validation.

This is already a rich research story: clinical extraction systems can look similar by aggregate F1 while differing radically in attribution, evidence validity, and generalization risk.

Closer To The Metric Goal

On validation, yes: we have reached or exceeded 0.9000 Purist on several development surfaces. But the metric goal is now correctly understood as conditional on attribution and generalization. A 0.9000 validation score is not enough if it is repair-heavy, validation-overfit, or candidate-recall-limited.

Closer To A Generalizing System

Moderately. The locked-test results remain sobering. The best frozen hybrid audit ties deterministic Purist on test rather than improving it. Claim-table v5 underperforms deterministic V1 on test but has complementary correct rows. The state graph is the best path toward generalization because it lets us measure which clinical states are represented before projection collapses them into Gan labels.

Research Thesis As Of Now

The strongest current thesis is not "LLMs beat rules" or "rules are enough." The strongest thesis is:

> Seizure-frequency extraction requires an auditable clinical-state > representation. Deterministic rules, LLM extraction, and benchmark > normalization can all improve performance, but they must be named by > prediction-bearing role. Generalization depends less on the final model family > label and more on whether the system preserves source-near evidence, > temporal/boundary semantics, and component-level attribution until final > projection.

This thesis is supported by the failures as much as by the successes.

Recommended Next Steps

1. Keep `rules_only_v1` and hybrid v0.2 `cluster_diary_candidate_recall` frozen. Do not tune from locked-test behavior.
2. Continue the state-graph cycle with a broader enriched-node regression panel before promoting duration projection policy.
3. Review `llm_heavy_clinical_frequency_reasoner_v1` validation250 failure families before another LLM-heavy prompt revision. Do not escalate v1 further: raw/format-only Purist is 188/250 and the stronger 219/250 selected-evidence arithmetic layer is attribution-diagnostic only.
4. Treat claim-table v5 as a comparator/complementarity source, not a promoted candidate, until it passes a fresh 25/50/250 ladder with written decision criteria.
5. Design LLM-replacement ablations over state-graph nodes: deterministic graph only, LLM boundary nodes only, LLM atomic claims, deterministic projection, LLM projection, and oracle projection.
6. Build hard panels by hidden subfamily: clusters, seizure-free duration, unknown/no-reference boundary, diary aggregation, recent-window arithmetic, competing semiologies, no-call/reparse repair sensitivity, and template family.
7. Keep every future aggregate score paired with attribution layers: raw, format-only, selected-evidence arithmetic, benchmark-aligned, deterministic fallback, and full stack where applicable.

Source Artifacts

Primary sources for this synthesis:

- `PROJECT_STATUS.md`
- `docs/research/contribution_thesis.md`
- `docs/design/architecture.md`
- `docs/design/gan2026_pipeline_v1.md`

- docs/research/gan2026_current_pipeline_results_report_2026-06-01.md
- experiments/gan2026_generalization_gap_research_report_2026-06-02.md
- experiments/gan2026_llm_structured_decision_retrospective_2026-06-01.md
- experiments/gan2026_section_claim_table_validation750_v4_interpretation_2026-06-01.md
- experiments/gan2026_architecture_validation250_component_ablation_2026-06-01.md
- experiments/gan2026_arch2_validation750_v01_interpretation_2026-06-01.md
- experiments/gan2026_hybrid_adjudicator_v02_selective_action_report_2026-06-01.md
- experiments/gan2026_hybrid_rules_candidates_llm_adjudicator_validation750_gpt41mini_v02_cluster_diary_candidate_recall_live_2026-06-02.md
- experiments/gan2026_hybrid_rules_candidates_llm_adjudicator_test450_gpt41mini_v02_cluster_diary_candidate_recall_live_2026-06-02.md
- experiments/gan2026_clinical_frequency_state_graph_protocol_2026-06-02.md
- experiments/gan2026_clinical_frequency_state_graph_validation50_diagnostics_2026-06-02.md
- experiments/gan2026_hybrid_clinical_frequency_state_graph_projection_arbitration_ablation_2026-06-02.md
- experiments/gan2026_hybrid_clinical_frequency_state_graph_seizure_free_duration_node_replay_2026-06-02.md
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- experiments/registry.jsonl
- experiments/RUN_INDEX.md